

Claim: Random Search needs with probability $1 - e^{-\Omega(n)}$ at least a budget of $2^{\frac{n}{2}}$ fitness evaluations to reach an optimal search point for the function F1.

Proof:

F1 is the OneMax problem. Random search will pick solutions uniformly from $\{0, 1\}^n$. The solution will be a n -digit binary string, with 2^n combinations. There is one optimal solution. Therefore, probability of getting optimal solution in one try is:

$$p = \frac{1}{2^n}$$

After T evaluations, probability of never sampling the optimum is (since the question asks for “at least” a budget, calculate lowerbound):

$$(1 - p)^T = \left(1 - \frac{1}{2^n}\right)^T$$

Choosing budget $T = 2^{\frac{n}{2}}$, then:

$$p[\text{no optimum in } T \text{ trials}] = \left(1 - \frac{1}{2^n}\right)^T, T = 2^{\frac{n}{2}}$$

But the above is extremely hard to deal with, so place it within a neat bound from the lecture:

$$\text{Upper bound: } 1 - x \leq e^{-x}, \quad x = \frac{1}{2^n}$$

$$\therefore \left(1 - \frac{1}{2^n}\right)^{2^{\frac{n}{2}}} \leq e^{-2^{-\frac{n}{2}}}$$

From the above inequality, we know that the probability of no optimum after T trials is at most

$$e^{-2^{-\frac{n}{2}}} \approx 1 - 2^{-\frac{n}{2}} \quad \left\{ \text{Taylor series where: } e^{-x} = 1 - x + \frac{x^2}{2} - \dots \right\}$$

Hence, after $2^{\frac{n}{2}}$ trials, random search will not reach optimum with probability of

$$1 - 2^{-\frac{n}{2}} \approx 1 - e^{-\Omega(n)},$$

proves that random search will need **at least** $2^{\frac{n}{2}}$ trials to reach optimum with probability $1 - e^{-\Omega(n)}$.

Claim: Prove that the number of fitness evaluations of RLS and (1+1) EA to reach an optimal search point for the function F3 is $O(n \log n)$ with probability $\Omega(1)$.

From the lecture slides we will be using fitness-based partition from Runtime Analysis 1, stating that we split search space into groups by fitness levels $A_0, A_1, A_2 \dots, A_n$, such that for the context of OneMax, A_1 has all solutions with one 1-bit, and A_n has all solutions with n 1-bits, etc. if p_i represents lower bound on the probability to leave level A_i upward in one iteration, then:

$$E[T] \leq \sum_{i=0}^{n-1} \frac{1}{p_i}, \quad T = \text{number of fitness evaluations}$$

And also Markov's Inequality:

$$\Pr(X \geq s \cdot E[X]) \leq \frac{1}{s}$$

Which can be flipped around also to bound the probability of at most that side:

$$\Pr(X \leq s \cdot E[X]) \geq 1 - \frac{1}{s}$$

For **RLS**:

At level i , RLS will flip exactly one bit at random. The probability of the flipped bit being 0-bit is:

$$p_i \geq \frac{n-i}{n}, \quad \text{since } i = \text{number of 1-bits}$$

Therefore, by fitness-partition bounds:

$$E[T_{RLS}] \leq \sum_{i=0}^{n-1} \frac{1}{p_i} \leq \sum_{i=0}^{n-1} \frac{n}{n-i} = n \sum_{j=1}^n \frac{1}{j} \quad [1]$$

The last sum we see there is the n^{th} harmonic number $H_n = \sum_{j=1}^n \frac{1}{j}$, and $H_n = \Theta(\log n) = O(n \log n)$.

We can convert the expectation to a constant probability statement via Markov's Inequality, where $s = 2$, since it keeps the probability clean:

$$\Pr(T_{RLS} \leq 2E[T_{RLS}]) \geq \frac{1}{2}$$

Thus $T_{RLS} = O(n \log n)$ with probability at least $\frac{1}{2}$, i.e., with probability $\Omega(1)$.

For **(1+1) EA**:

For (1+1) EA, each bit will be flipped with chance of $\frac{1}{n}$, the probability of not getting flipped is therefore $1 - \frac{1}{n}$. So at level i , the probability that we flip one specific 0-bit and keep all other bits the same is:

$$\frac{1}{n} \cdot \left(1 - \frac{1}{n}\right)^{n-1} \geq \frac{1}{e \cdot n}$$

And at level i , there are $n - i$ 0-bits that can be flipped into 1-bits, so

$$p_i \geq \frac{n - i}{e \cdot n}$$

Therefore, exactly as the fitness-partitions shown in Runtime Analysis 1 slide 23,

$$\begin{aligned} E[T_{(1+1)EA}] &\leq \sum_{i=0}^{n-1} \frac{1}{p_i} \leq \sum_{i=0}^{n-1} \frac{e \cdot n}{n - i} \leq e \cdot n \sum_{i=0}^{n-1} \frac{1}{n - i} \\ &= e \cdot n \sum_{j=1}^n \frac{1}{j} = O(n \log n) \quad [1] \end{aligned}$$

We can once again apply Markov's inequality with $s = 2$

$$\Pr(T_{(1+1)EA} \leq 2 E[T_{(1+1)EA}]) \geq \frac{1}{2},$$

So $T_{(1+1)EA} = O(n \log n)$ with probability $\Omega(1)$.

[1]

Letting $j = n - i$,

$$\sum_{i=0}^{n-1} \frac{1}{n - i} = \sum_{j=n}^1 \frac{1}{j}$$

Since when $i = 0 \rightarrow j = n$, and when $i = n - 1 \rightarrow j = 1$. As i increases, j decreases. We can then reverse the sum since the order of the summation doesn't matter.

$$\sum_{j=n}^1 \frac{1}{j} = \sum_{j=1}^n \frac{1}{j}$$